

XIAMEN UNIVERSITY

Programming Language C

Week 3 - Chapter 2

Types, Operators and Expressions (ii)

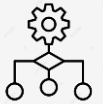
Lab

Dr. Najla'a

Outline

- 2.1 Basic C Data Types
- 2.2 Fundamental Operators
- 2.3 Operator Precedence
- 2.4 Expressions and Statements

Type Casting and Floating-Point Division

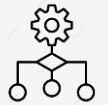


```c
#include <stdio.h>
#include <stdlib.h>
//Ex3
int main()
{
    int a = 7, b = 2;
    float result = a / b;
    printf("Without casting: %.2f\n", result);

    result = (float)a / b;
    printf("With casting: %.2f\n", result);

}
```

2.2.2 Relational operators



- E.x.1: Using relational operators, and equality operators, write a simple C program that compares two integers entered by the user and prints whether they are equal or not?

- **Note**: do not use “if” statement.

2.2.2 Relational operators

- E.x.1: Solution

```c
#include <stdio.h>

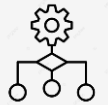
int main() {
    int num1, num2;

    // Input two integers from the user
    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    // Directly display the result of the condition
    printf("The numbers are %s\n", (num1 == num2) ? "equal" : "not equal");


    return 0;
}
```

2.2.3 Logical operators



- Ex2: Write a program to check if a number is within a given range using the conditional operator.

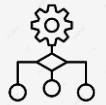
2.2.3 Logical operators

- Ex2: Solution

```c
#include <stdio.h>

int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    printf("%s\n",(num >= 10 && num <= 20) ? "Within range" : "Out of range");
}
```

2.2.6 Conditional operator



- **Ex4:**Write a program that takes three integers as input and prints the largest of the three numbers using the conditional (ternary) operator. Additionally, if all the numbers are equal, the program should print a message indicating that they are equal.

2.2.6 Conditional operator

- **Ex4:** Solution

```c
#include <stdio.h>

int main() {
    int a, b, c, largest;

    // Input three integers
    printf("Enter three integers: ");
    scanf("%d %d %d", &a, &b, &c);

    // Find the largest number using nested conditional operators
    largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);

    // Print the largest number
    printf("The largest number is: %d\n", largest);

    // Check if all numbers are equal using the conditional operator
    printf("%s\n", (a == b && b == c) ? "All numbers are equal" : "The numbers are not equal");

    return 0;
}
```

Practice

- Ex.5 Write a program that reads an integer and determines and prints whether it's odd or even. [Hint: Use the remainder operator. An even number is a multiple of two. Any multiple of two leaves a remainder of zero when divided by 2.]

Practice

- Ex.6 Write a program that asks the user to enter the highest rainfall ever in one season for a country, and the rainfall in the current year for that country, obtains the values from the user, checks if the current rainfall exceed the highest rainfall and prints an appropriate message on the screen. If the current rainfall is higher, it assigns that value as the highest rainfall ever. Use only the single-selection form of the if statement you learned in this chapter.

Practice

- Ex.6 solution

```c
#include <stdio.h>

int main() {
    float highest_rainfall, current_rainfall;

    // Ask the user to enter the highest rainfall ever recorded
    printf("Enter the highest rainfall ever recorded (in mm): ");
    scanf("%f", &highest_rainfall);

    // Ask the user to enter the rainfall for the current year
    printf("Enter the current year's rainfall (in mm): ");
    scanf("%f", &current_rainfall);

    // Use the conditional operator to check and update the highest rainfall
    highest_rainfall = (current_rainfall > highest_rainfall) ? current_rainfall : highest_rainfall;

    // Print the message if the current rainfall exceeded the highest rainfall
    printf("%s\n", (current_rainfall > highest_rainfall) ? "This year's rainfall exceeded the highest recorded rainfall!" : "");

    // Print the current highest rainfall
    printf("The highest rainfall recorded is: %.2f mm\n", highest_rainfall);

    return 0;
}
```

Practice

- Ex.7 Using only the techniques you learned in this chapter, write a program that calculates the squares and cubes of the numbers from 0 to 10 and uses tabs to print the following table of values:

number	square	cube
0	0	0
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000

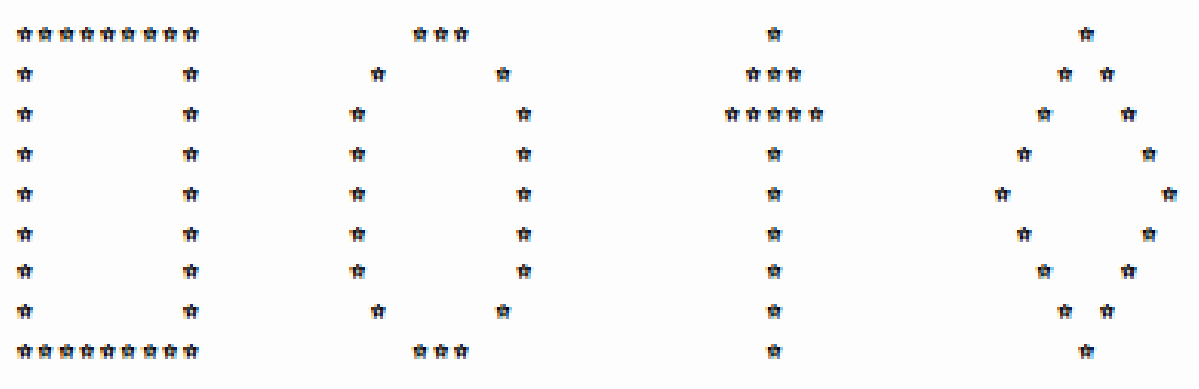
Practice

- Ex.7. Solution

```
#include<stdio.h>
int main(void)
{
printf("Number  Square  Cube\n");
printf("0\t0\t0\n");
printf("1\t1\t1\n");
printf("2\t4\t8\n");
printf("3\t9\t27\n");
printf("4\t16\t64\n");
printf("5\t25\t125\n");
printf("6\t36\t216\n");
printf("7\t49\t343\n");
printf("8\t64\t512\n");
printf("9\t81\t729\n");
printf("10\t100\t1000\n");
}
```

Practice

- Ex.8 Write a program that prints the following shapes with asterisks.



Practice

- Ex.8. Solution

```
#include<stdio.h>
int main(void)
{
printf("*********          ***              *              *\n");
printf("*        *        *   *           ***            * *\n");
printf("*        *       *     *         *****          *   *\n");
printf("*        *       *     *           *           *     *\n");
printf("*        *       *     *           *         *       *\n");
printf("*        *       *     *           *           *     *\n");
printf("*        *       *     *           *            *   *\n");
printf("*        *        *   *            *             * * \n");
printf("*********          ***              *              *\n");
}
```

Practice

- Ex.9 Write a program that prints a shape in your brain with asterisks. For example, print “LOVE” with asterisks.

Practice

- Ex.9. Solution

```
#include<stdio.h>
int main(void)
{
printf("*\t\t**\t\t*\t*\t\t*********\n*");
printf("\t      *\t     *\t\t*\t*\t\t*\n*");
printf("\t     *\t      *\t\t *      *\t\t*********\n*");
printf("\t      *\t     *\t\t *      *\t\t*\n");
printf("*******\t\t**\t\t     *\t\t\t*********");
}
```

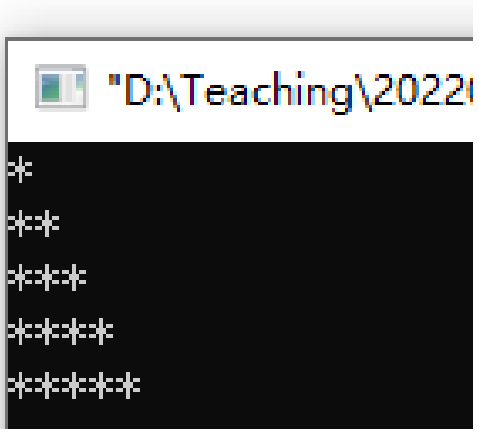
Practice

- Ex.10 What does the following statement print?
  - printf("*\n**\n***\n****\n*****\n");

Practice

- Ex.10. Solution

```
#include<stdio.h>
int main(void)
{
printf( "*\n**\n***\n****\n*****\n" );
}
```





Thank you

Q&A